

PRATIK SARODE

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UK WORK ELIGIBILITY | UK GRADUATE ROUTE VISA | FULL RIGHT TO WORK UNTIL JAN 2028

SUMMARY

AI Engineer with **2+ years** of experience building low-latency, multi-agent systems and advanced RAG pipelines. Backed by an MSc in Applied AI, I leverage frameworks like LangChain, LangGraph, and CrewAI to design production-ready, cloud-based LLM applications. Adept at driving complex automation projects from end to end, with a strong track record of optimizing system performance and engineering robust hallucination guardrails.

TECHNICAL SKILLS

Languages: Python, Next.js, JavaScript/TypeScript

AI/ML: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Prompt Engineering, AI Agents, LangGraph, MCP, Agentic AI

Frameworks: TensorFlow, PyTorch, scikit-learn, LangChain, LangGraph, Hugging Face, FastAPI, Streamlit, Gradio

Backend: FastAPI, REST APIs, API Integration, Backend Development, Database Models, CRUD Operations, Authentication

Data/Vector Search: Pandas, NumPy, FAISS, Qdrant, PostgreSQL, Embeddings, Semantic Search

Tools: Git, GitHub, Docker, MLflow, Jupyter Notebook, Technical Documentation

EDUCATION

MSc Applied Artificial Intelligence and Data Analytics (First Class) <i>University of Bradford, UK</i>	<i>Sept 2024 - Sept 2025</i>
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BE. Electronics and Telecommunications (Distinction) GPA-7.76 <i>AISSMS Institute of Information and Technology</i>	<i>Jan 2021 - June 2024</i>
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EXPERIENCE

AI Engineer Gamaka Intellectual Machines	<i>Oct 2024 - Present</i>
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- Spearheaded the development of real-world Generative AI and Agentic AI applications, functioning as a core engineer for a Multi-Agent Autonomous Medical Insurance Claim Validation Engine.
- Architected advanced RAG pipelines utilizing LangChain and Langflow to optimize data retrieval, improve LLM accuracy, and ground enterprise responses in proprietary data.
- Orchestrated multi-agent workflows using CrewAI, successfully implementing the Model Context Protocol (MCP) to maintain seamless state and context management across autonomous agents.
- Engineered sophisticated prompts and integrated the OpenAI API to build robust, conversational AI interfaces tailored for complex financial domain use cases.
- Bridged healthcare and finance domains by deploying production-ready AI solutions that significantly reduced manual validation time and improved overall conversational accuracy.

CERTIFICATIONS

- Oracle Cloud Infrastructure Gen AI Professional
- Kaggle Python Coder
- AI & Agile (Scrum Alliance)
- AI Ethics & Agile Project Leadership (Excellidia)

ACHIEVEMENTS

- Open-source contributor to ElizaOS DeFi Oracle Agent plugin for Nosana x ElizaOS Challenge
- Student Representative at University of Bradford
- Top rank in Excellidia's AI Leadership program (50+ participants)

KEY PROJECTS

Autonomous Medical Insurance Claim Validation Engine | Python, LangGraph, FastAPI, PostgreSQL *Gamaka Intellectual Machines*

- Architected a production-oriented multi-agent clinical validation engine using LangGraph, LLMs, Pydantic v2, and asynchronous Python, orchestrating six domain-specific validation agents in parallel after dependency-aware ICD coding, with typed outputs, checkpointing, fault tolerance, and human-in-the-loop medical review.
- Engineered a Hybrid RAG pipeline for ICD-10 coding, combining BM25 sparse retrieval, semantic embeddings, Reciprocal Rank Fusion (RRF), FAISS/pgvector, and constrained LLM selection to retrieve and rank the most relevant diagnosis codes while reducing unsupported code generation and improving retrieval precision across clinical shorthand and paraphrased diagnoses.
- Designed a clinical evidence-grounding framework enforcing mandatory verbatim quote verification against source documents before any narrative finding could be accepted; implemented fuzzy matching with RapidFuzz strictly for diagnostics/debugging while retaining exact-match verification as the production hallucination gate.
- Implemented structured AI outputs and automated abstention mechanisms using Pydantic schemas and validation gates, enabling the system to return ABSTAIN for ambiguous, high-risk, insufficient-evidence, or statistically unreliable cases and route them to human medical officers rather than generating unsupported conclusions.
- Built a version-controlled clinical knowledge base using PostgreSQL with bitemporal versioning, Git-backed governance, CI validation, and medical-advisor review, enabling historical reconstruction of diagnosis-drug, diagnosis-procedure, investigation, day-care, and PED mappings for auditability and reproducibility.
- Developed a scalable async FastAPI serving layer integrated with a Streamlit medical-officer review dashboard, supporting concurrent LLM execution, structured API contracts, real-time evidence inspection, and human approval/query/denial workflows; designed production deployment across GCP Cloud Run and GKE for stateless APIs and stateful orchestration workers.
- Established an evaluation and observability layer using Langfuse, pytest, property-based testing, and automated edge-case validation, capturing prompts, model responses, retrieved evidence, latency, token usage, and validation outcomes to support reproducible testing, debugging, and compliance-oriented auditing.
- Optimized inference efficiency through semantic caching, prompt caching, retrieval optimization, and parallel agent execution, targeting substantial reductions in redundant LLM calls, token consumption, and end-to-end validation latency while maintaining evidence-grounded decision quality.

Agentic AI System for Fraud & AML Risk Triage | Python, LangGraph, scikit-learn, FAISS *Gamaka Intellectual Machines*

- Trained and tuned statistical anomaly-detection and graph-based network-analysis models (scikit-learn) on transaction data to score fraud/AML risk, forming the deterministic ML tier that pre-screens alerts before any LLM involvement.
- Architected a LangGraph-powered multi-agent triage pipeline that routes only ML-flagged high-risk alerts to LLM review, resolving the majority of alerts via the ML/rules tier alone and capping expensive LLM usage to exactly two calls per escalated case.
- Engineered a RAG-grounded Suspicious Activity Report (SAR) drafting engine that cites every factual claim directly to source text spans for auditable transparency.
- Implemented a dual-agent adversarial validation loop, using a secondary LLM to independently verify drafted claims before any statement reaches a human analyst.
- Enforced a typed schema with mandatory abstention and fallback logic, so the system defers to human review rather than acting on low-confidence signals.
- Established full audit-trail provenance across the ML and LLM tiers, aligning the architecture with model-risk documentation standards used in BFSI compliance.